

# Action-Type-Conditioned Grounding for GUI Agents

Aadi Chauhan  
aadic@stanford.edu

Arthur Ilyasov  
arthuri@stanford.edu

Nevin Kunampuram  
nevinkun@stanford.edu

April 24, 2026

## 1 Proposal

Vision-language GUI agents must, at every step, make two coupled decisions: what kind of action to perform (click, double-click, type, scroll, drag, hotkey) and where on the screen to perform it. Current state-of-the-art models including UI-TARS [7], OS-Atlas [10], and SeeClick [2] entangle these decisions in a single autoregressive decode, emitting action-type tokens and spatial coordinate tokens sequentially from the same head. This entanglement creates a failure mode observed across deployed systems: the model commits to an incorrect action type — for example, scrolling past a visible target instead of clicking it — and then grounds coherently but wrongly within that incorrect type. We investigate whether explicitly decoupling action-type prediction from spatial grounding — predicting the action type first via a dedicated head, then conditioning the grounding stage on the predicted type via a learned action-type embedding — improves both action-type selection and downstream grounding accuracy. This tests a specific hypothesis about the benefits of structured decomposition over flat autoregressive decoding in multimodal agents, and action-type errors compound catastrophically in real-world agentic deployments.

We will draw on prior work in GUI agent modeling [7, 10, 2], conditional computation in vision-language models, and LoRA-based parameter-efficient fine-tuning [5]. Our primary data is Mind2Web [3] (~137k labeled web steps), supplemented with Android in the Wild [8] (~715k mobile episodes) and AndroidControl [6], all providing aligned (screenshot, instruction, action-type, target) tuples. We will unify these sources under a canonical 8-class action taxonomy and apply class-balanced sampling to address the heavy click-majority imbalance.

Our method is a two-stage pipeline. Stage one is an action-type classifier: a 3-layer MLP on pooled visual and instruction embeddings from a frozen Qwen2-VL-7B [9], trained with class-weighted cross-entropy. Stage two is action-type-conditioned grounding: a learned embedding per action class is prepended to the instruction tokens, and the VLM is LoRA-fine-tuned on Wave-UI-25K [1] and Mind2Web grounding pairs with the predicted type as a conditioning signal. Evaluation will report action-type classification F1 on held-out Mind2Web, Showdown Clicks [4] top-1 accuracy to probe grounding improvements, and Mind2Web step success rate to probe joint performance. Ablations will compare flat joint decoding, auxiliary action-type loss only, hard constrained decoding, and our conditioning-embedding variant. Qualitative analysis will visualize attention shifts induced by action-type conditioning.

## 2 References

- [1] AgentSea. *Wave-UI-25K: A Dataset for Fine-Tuning Web Interaction Agents*. 2024. URL: <https://huggingface.co/datasets/agentsea/wave-ui-25k> (visited on 04/23/2026).
- [2] Kanzhi Cheng et al. “SeeClick: Harnessing GUI Grounding for Advanced Visual GUI Agents”. In: *arXiv preprint arXiv:2401.10935* (2024).
- [3] Xiang Deng et al. “Mind2Web: Towards a Generalist Agent for the Web”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023.
- [4] General Agents. *The Showdown Computer Control Evaluation Suite*. 2025. URL: <https://github.com/generalagents/showdown> (visited on 04/23/2026).
- [5] Edward J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models”. In: *International Conference on Learning Representations (ICLR)*. 2022.
- [6] Wei Li et al. “On the Effects of Data Scale on UI Control Agents”. In: *arXiv preprint arXiv:2406.03679* (2024).
- [7] Yujia Qin et al. “UI-TARS: Pioneering Automated GUI Interaction with Native Agents”. In: *arXiv preprint arXiv:2501.12326* (2025).
- [8] Christopher Rawles et al. “Android in the Wild: A Large-Scale Dataset for Android Device Control”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023.
- [9] Peng Wang et al. “Qwen2-VL: Enhancing Vision-Language Model’s Perception of the World at Any Resolution”. In: *arXiv preprint arXiv:2409.12191* (2024).
- [10] Zhiyong Wu et al. “OS-Atlas: A Foundation Action Model for Generalist GUI Agents”. In: *arXiv preprint arXiv:2410.23218* (2024).